

CYBR 437 Secure Coding
Fall 2026
Lab - 2 GDB Primer
Release date: October 1, 2026
Due date: October 7, 2026, at 11:59 pm

Introduction

You will use GDB to analyze the given binary file *lab2* and answer the following questions. The source files are also included (*sfiles*). If compiling from the source files, be sure to use the `-g` option to include debugging info.

1. (5 points) What type of file is this, and what kind of security does it have?
2. (5 points) What are the files that make up this binary, and which one contains main?
3. (5 points) What are the type(s) and name(s) of parameter(s) being passed to main?
4. (5 points) Set a breakpoint on each function and display the breakpoints.
5. (5 points) Begin running the program. How many arguments are passed to main?
6. (5 points) Step twice and display the contents of the constant and two variables in main.
7. (5 points) Step into the first function called in main. What is the type and name of the constant? Where is the constant declared?
8. (5 points) Explain the difference between printing `var` and `var[x]`?
9. (5 points) Instead of continuing this function, return to main and print the current line.
10. (5 points) Step into the second function called from main. What is the name and starting value of the counting variable?
11. (5 points) Use the disassemble command in GDB to display the assembly code for the function. What does the output show, and how does it correlate with the C source code?
12. (5 points) Delete the breakpoint for the first function called in main and disable the breakpoint for the last function called.

13. (5 points) Without starting over, AKA your current location in a function that is not main, print the memory location of the first variable passed to main.
14. (5 points) Print the information on the current running threads. How many threads are running?
15. (5 points) Enable the breakpoint for the third function called by main. What is the type, name, and memory location of the variable passed to it?

Submission Instructions: Please submit a single PDF file in Canvas that includes the correct GDB command, as applicable, and screen capture(s) of the output.